CSE 708
Graph Theory

Name: Md. Samiel Islam Sami
Id: 21301002

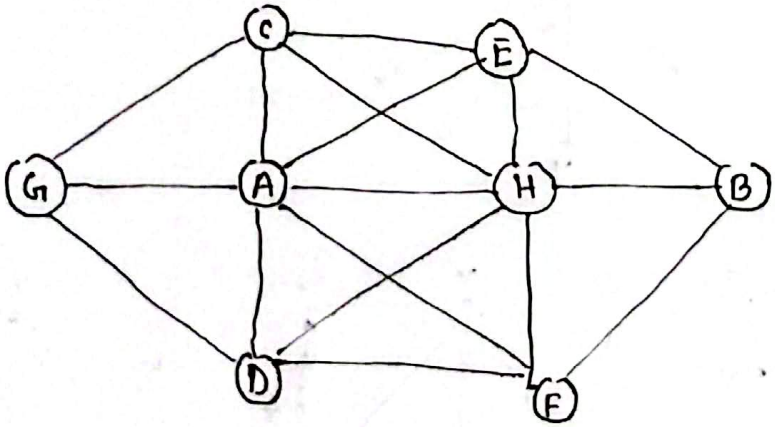
Three Puzzles :

(1) The Eight Circles Problem

Place the letters A, B, C, D, E, F, G, H into eight circles so that no letter is adjacent to a letter immediately before or after in the alphabet.

i) the easiest letter to place are A and H as only B and G respectively can not be adjacent. So we place A and H in the middle class circles.

ii) The letter C must be now placed on the left side of the graph and F must be placed on the right side of the graph. After that we place the remaining alphabets according to the rule.



(2) Six People at a Party:

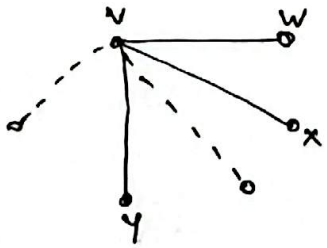
Show that in any gathering of six people, there are either three people who all know each either or three people none of whom co'knows of the other two.

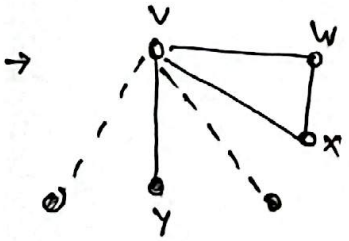
Which means in six group people always contains

- three mutual friends
- three mutual stanranges

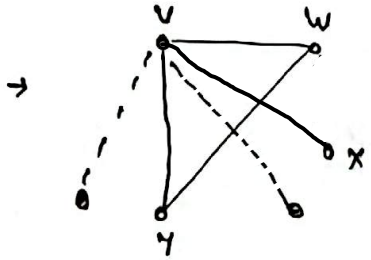
Here each person is represented by vertices and edges represent their relationship. Solid edge corresponds if they know each other. Dotted edge represent if they are strangers.

Let 'v' be any vertex. There must be 5 edges, either three solid on dotted.

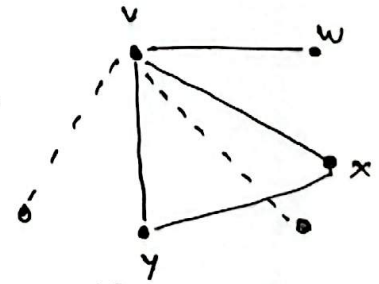




if w and x know each other

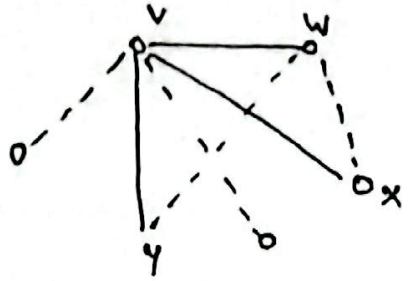


if w and y know each other

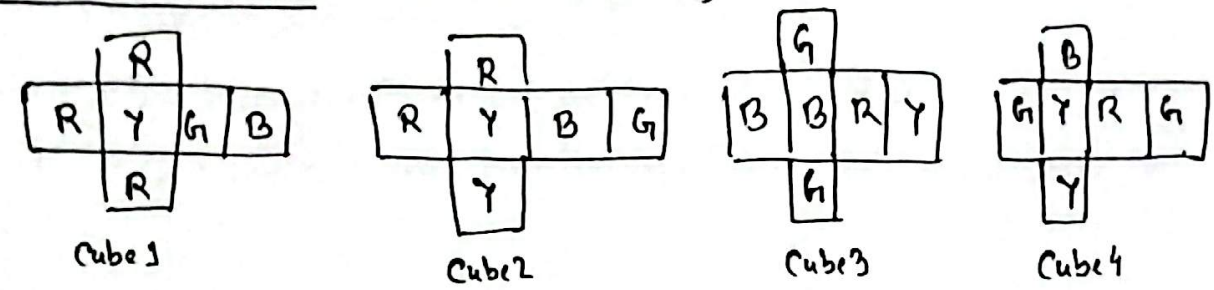


if x and y know each other

Finally if no two people of w, x and y know each other w, x, y form a dotted triangle

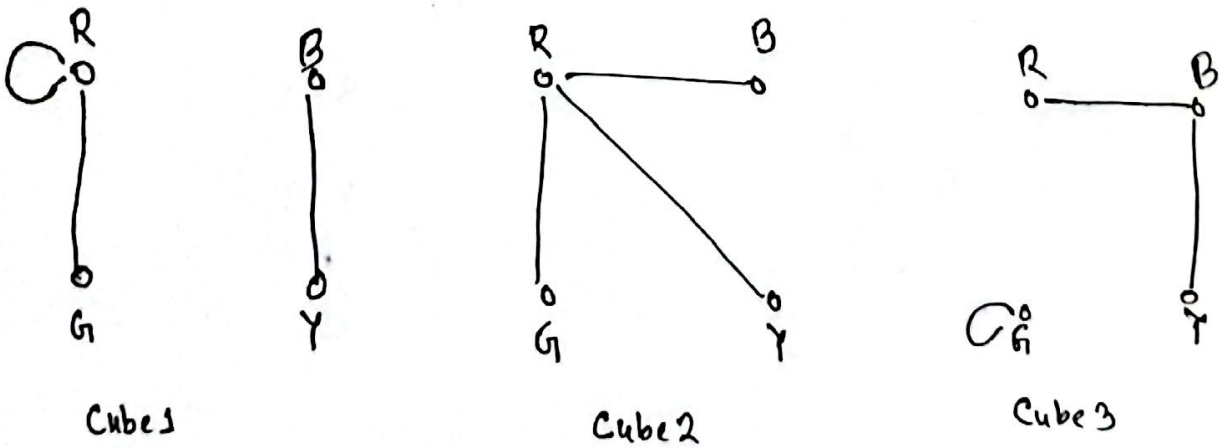


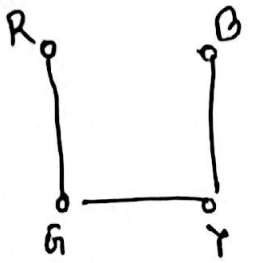
③ Four Cubes Problem (Instant Insanity)



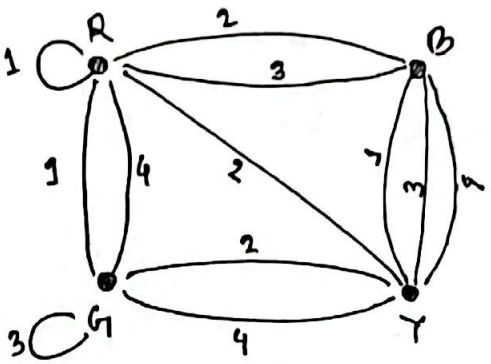
The task is to stack the four cubes into a vertical tower so that: the front, back, left, right side contains all four colors exactly once. The top and bottom face color doesn't matter.

This is solved by representing the cubes in graph with four vertices R, B, G, Y for each colors. In the graph two vertices are adjacent if and only if the cube has corresponding color on the opposite side.



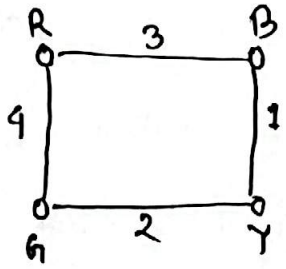


cube4

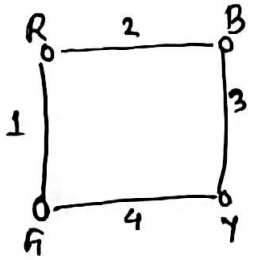


Superimposed Graph

The number in superimposed graph's edges signifies the cube number. From the superimposed graph we can create two subgraph H1 and H2. H1 tells us which color appears on the front and back. Meanwhile, H2 tells us which color appears on the left and right side.

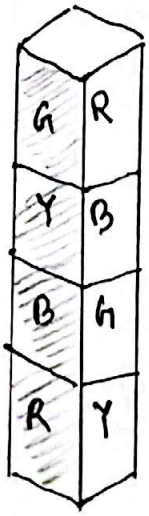


front and back
H1

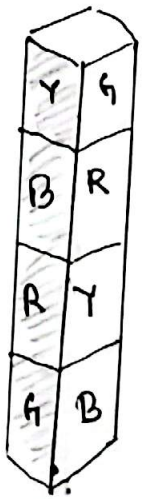


left and Right
H_2

from the graph H1 and H_2 we can get the proper cube stacking order.



left front



right back